```
Windows PowerShell
PS D:\DTCC\GUVI\DevOps> mkdir python-da


    Directory: D:\DTCC\GUVI\DevOps


Mode                 LastWriteTime         Length Name
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d-----        26-10-2026     15:53                python-da


PS D:\DTCC\GUVI\DevOps> cd .\python-da\
PS D:\DTCC\GUVI\DevOps\python-da> py -m venv .venv
PS D:\DTCC\GUVI\DevOps\python-da> .\.venv\Scripts\Activate.ps1
(.venv) PS D:\DTCC\GUVI\DevOps\python-da> .\.venv\Scripts\python.exe --version
Python 3.12.1
(.venv) PS D:\DTCC\GUVI\DevOps\python-da> python -m pip install --upgrade pip
Requirement already satisfied: pip in d:\dtcc\guvi\devops\python-da\.venv\lib\site-packages (23.2.1)
Collecting pip
  Obtaining dependency information for pip from https://files.pythonhosted.org/packages/3f/06/3e4420302363fbb71be3c7f35ce042f54bfca330e3e66b3a2e3c4cca579f/pip-26.2.2-py3-none-any.whl.metadata
  Downloading pip-26.2.2-py3-none-any.whl.metadata (4.6 kB)
Downloading pip-26.2.2-py3-none-any.whl (1.8 MB)
   ---------------------------------------- 1.8/1.8 MB 2.1 MB/s eta 0:00:00
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 23.2.1
    Uninstalling pip-23.2.1:
      Successfully uninstalled pip-23.2.1
Successfully installed pip-26.2.2
(.venv) PS D:\DTCC\GUVI\DevOps\python-da> python -m pip install jupyterlab numpy pandas
Collecting jupyterlab
  Downloading jupyterlab-4.6.4-py3-none-any.whl.metadata (16 kB)
Collecting numpy
  Downloading numpy-2.5.4-cp312-cp312-win_amd64.whl.metadata (6.6 kB)
Collecting pandas
  Downloading pandas-3.1.0-cp312-cp312-win_amd64.whl.metadata (18 kB)
Collecting async-lru>=1.0.0 (from jupyterlab)
  Downloading async_lru-2.4.0-py3-none-any.whl.metadata (10 kB)
Collecting httpx<1,>=0.25.0 (from jupyterlab)
  Using cached httpx-0.28.1-py3-none-any.whl.metadata (7.1 kB)
Collecting ipykernel!=6.30.0,>=6.5.0 (from jupyterlab)
  Downloading ipykernel-7.4.0-py3-none-any.whl.metadata (3.4 kB)
Collecting jinja2>=3.0.3 (from jupyterlab)
  Using cached jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting jupyter-builder>=1.0.2 (from jupyterlab)
  Downloading jupyter_builder-1.2.3-py3-none-any.whl.metadata (7.8 kB)
```

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Windows PowerShell
Downloading executing-2.3.0-py3-none-any.whl (29 kB)
Downloading terminado-0.18.1-py3-none-any.whl (14 kB)
Downloading tinycss2-1.5.1-py3-none-any.whl (28 kB)
Downloading tornado-6.6-cp311-abi3-win_amd64.whl (530 kB)
   ---------------------------------------- 530.9/530.9 kB 8.7 MB/s  0:00:00
Downloading traitlets-5.16.1-py3-none-any.whl (86 kB)
Downloading typing_extensions-4.16.0-py3-none-any.whl (45 kB)
Downloading wcwidth-0.9.4-cp310-abi3-win_amd64.whl (597 kB)
   ---------------------------------------- 597.8/597.8 kB 22.4 MB/s  0:00:00
Downloading webcolors-25.10.0-py3-none-any.whl (14 kB)
Downloading webencodings-0.6.1-py3-none-any.whl (8.7 kB)
Downloading websocket_client-1.9.2-py3-none-any.whl (95 kB)
Downloading argon2_cffi_bindings-26.1.0-cp310-abi3-win_amd64.whl (25 kB)
Using cached cffi-2.1.1-cp312-cp312-win_amd64.whl (185 kB)
Downloading beautifulsoup4-4.15.0-py3-none-any.whl (109 kB)
Downloading soupsieve-3.0.4-py3-none-any.whl (42 kB)
Downloading defusedxml-0.7.1-py2.py3-none-any.whl (25 kB)
Downloading fqdn-1.6.0-py3-none-any.whl (12 kB)
Downloading isoduration-20.11.0-py3-none-any.whl (11 kB)
Downloading arrow-1.4.0-py3-none-any.whl (68 kB)
Downloading jupyterlab_pygments-0.3.0-py3-none-any.whl (15 kB)
Downloading pure_eval-0.2.4-py3-none-any.whl (11 kB)
Downloading pycparser-3.12-py3-none-any.whl (62 kB)
Downloading rfc3339_validator-0.1.4-py2.py3-none-any.whl (3.5 kB)
Downloading tzdata-2026.5-py2.py3-none-any.whl (347 kB)
Downloading uri_template-1.3.0-py3-none-any.whl (11 kB)
Installing collected packages: pure-eval, websocket-client, webencodings, webcolors, wcwidth, urllib3, uri-template, tzdata, typing_extensions, traitlets, tornado, soupsieve, six, send2trash, rpds-py, rfc3986-validator, pyzmq, pyyaml, pywinpty, python-json-logger, pygments, pycparser, psutil, prometheus-client, platformdirs, parso, pandocfilters, packaging, numpy, nest-asyncio2, mistune, MarkupSafe, lark, jupyterlab-pygments, jsonpointer, json5, idna, h11, fqdn, fastjsonschema, executing, defusedxml, debugpy, comm, colorama, charset_normalizer, certifi, babel, attrs, async-lru, asttokens, tinycss2, terminado, stack_data, rfc3987-syntax, rfc3339-validator, requests, referencing, python-dateutil, prompt_toolkit, matplotlib-inline, jupyter-core, jinja2, jedi, ipython-pygments-lexers, httpcore, cffi, bleach, beautifulsoup4, anyio, pandas, jupyter-server-terminals, jupyter-client, jupyter-builder, jsonschema-specifications, ipython, httpx, arrow, argon2-cffi-bindings, jsonschema, isoduration, ipykernel, argon2-cffi, nbformat, nbclient, jupyter-events, nbconvert, jupyter-server, notebook-shim, jupyterlab-server, jupyter-lsp, jupyterlab
Successfully installed MarkupSafe-3.0.4 anyio-4.15.1 argon2-cffi-25.1.0 argon2-cffi-bindings-26.1.0 arrow-1.4.0 asttokens-3.0.2 async-lru-2.4.0 attrs-26.1.0 babel-2.18.0 beautifulsoup4-4.15.0 bleach-6.4.0 certifi-2026.10.23 cffi-2.1.1 charset_normalizer-3.5.2 colorama-0.4.6 comm-0.2.3 debugpy-1.8.22 defusedxml-0.7.1 executing-2.3.0 fastjsonschema-2.22.2 fqdn-1.6.0 h11-0.16.0 httpcore-1.0.9 httpx-0.28.1 idna-3.20 ipykernel-7.4.0 ipython-9.17.1 ipython-pygments-lexers-1.1.1 isoduration-20.11.0 jedi-0.20.1 jinja2-3.1.6 json5-0.17.3 jsonpointer-3.5 jsonschema-4.26.0 jsonschema-specifications-2025.9.1 jupyter-builder-1.2.3 jupyter-client-8.10.0 jupyter-core-5.9.1 jupyter-events-0.12.1 jupyter-lsp-2.3.1 jupyter-server-2.21.1 jupyter-server-terminals-0.5.4 jupyterlab-4.6.4 jupyterlab-pygments-0.3.0 jupyterlab-server-2.28.1 lark-1.3.1 matplotlib-inline-0.2.2 mistune-3.4.0 nbclient-0.11.0 nbconvert-7.17.2 nbformat-5.11.1 nest-asyncio2-1.7.4 notebook-shim-0.2.4 numpy-2.5.4 packaging-26.3 pandas-3.1.0 pandocfilters-1.5.1 parso-0.8.7 platformdirs-4.13.5 prometheus-client-0.26.0 prompt_toolkit-3.0.53 psutil-7.2.2 pure-eval-0.2.4 pycparser-3.12 pygments-2.21.0 python-dateutil-3.0.0 python-json-logger-4.2.0 pywinpty-3.1.0 pyyaml-6.0.3 pyzmq-27.2.0 referencing-0.37.0 requests-2.34.2 rfc3339-validator-0.1.4 rfc3986-validator-0.1.1 rfc3987-syntax-1.1.0 rpds-py-2026.9.1 send2trash-2.1.0 six-1.17.0 soupsieve-3.0.4 stack_data-0.6.3 terminado-0.18.1 tinycss2-1.5.1 tornado-6.6 traitlets-5.16.1 typing_extensions-4.16.0 tzdata-2026.5 uri-template-1.3.0 urllib3-2.8.0 wcwidth-0.9.4 webcolors-25.10.0 webencodings-0.6.1 websocket-client-1.9.2
(.venv) PS D:\DTCC\GUVI\DevOps\python-da>
```

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Windows PowerShell
Downloading typing_extensions-4.16.0-py3-none-any.whl (45 kB)
Downloading wcwidth-0.9.4-cp310-abi3-win_amd64.whl (597 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 597.8/597.8 kB 22.4 MB/s  0:00:00
Downloading webcolors-25.10.0-py3-none-any.whl (14 kB)
Downloading webencodings-0.6.1-py3-none-any.whl (8.7 kB)
Downloading websocket_client-1.9.2-py3-none-any.whl (95 kB)
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Using cached cffi-2.1.1-cp312-cp312-win_amd64.whl (185 kB)
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Downloading soupsieve-3.0.4-py3-none-any.whl (42 kB)
Downloading defusedxml-0.7.1-py2.py3-none-any.whl (25 kB)
Downloading fqdn-1.6.0-py3-none-any.whl (12 kB)
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Downloading pure_eval-0.2.4-py3-none-any.whl (11 kB)
Downloading pycparser-3.12-py3-none-any.whl (62 kB)
Downloading rfc3339_validator-0.1.4-py2.py3-none-any.whl (3.5 kB)
Downloading tzdata-2026.5-py2.py3-none-any.whl (347 kB)
Downloading uri_template-1.3.0-py3-none-any.whl (11 kB)
Installing collected packages: pure-eval, websocket-client, webencodings, webcolors, wcwidth, urllib3, uri-template, tzdata, typing_extensions, traitlets, t
ornado, soupsieve, six, send2trash, rpds-py, rfc3986-validator, pyzmq, pyyaml, pywinpty, python-json-logger, pygments, pycparser, psutil, prometheus-client,
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nts-lexers, httpcore, cffi, bleach, beautifulsoup4, anyio, pandas, jupyter-server-terminals, jupyter-client, jupyter-builder, jsonschema-specifications, ipy
thon, httpx, arrow, argon2-cffi-bindings, jsonschema, isoduration, ipykernel, argon2-cffi, nbformat, nbclient, jupyter-events, nbconvert, jupyter-server, no
tebook-shim, jupyterlab-server, jupyter-lsp, jupyterlab
Successfully installed MarkupSafe-3.0.4 anyio-4.15.1 argon2-cffi-25.1.0 argon2-cffi-bindings-26.1.0 arrow-1.4.0 asttokens-3.0.2 async-lru-2.4.0 attrs-26.1.0
 babel-2.18.0 beautifulsoup4-4.15.0 bleach-6.4.0 certifi-2026.10.23 cffi-2.1.1 charset_normalizer-3.5.2 colorama-0.4.6 comm-0.2.3 debugpy-1.8.22 defusedxml-
0.7.1 executing-2.3.0 fastjsonschema-2.22.2 fqdn-1.6.0 h11-0.16.0 httpcore-1.0.9 httpx-0.28.1 idna-3.20 ipykernel-7.4.0 ipython-9.17.1 ipython-pygments-lexe
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yterlab-pygments-0.3.0 jupyterlab-server-2.28.1 lark-1.3.1 matplotlib-inline-0.2.2 mistune-3.4.0 nbclient-0.11.0 nbconvert-7.17.2 nbformat-5.11.1 nest-async
io2-1.7.4 notebook-shim-0.2.4 numpy-2.5.4 packaging-26.3 pandas-3.1.0 pandocfilters-1.5.1 parso-0.8.7 platformdirs-4.13.5 prometheus-client-0.26.0 prompt_to
olkit-3.0.53 psutil-7.2.2 pure-eval-0.2.4 pycparser-3.12 pygments-2.21.0 python-dateutil-3.0.0 python-json-logger-4.2.0 pywinpty-3.1.0 pyyaml-6.0.3 pyzmq-27
.2.0 referencing-0.37.0 requests-2.34.2 rfc3339-validator-0.1.4 rfc3986-validator-0.1.1 rfc3987-syntax-1.1.0 rpds-py-2026.9.1 send2trash-2.1.0 six-1.17.0 so
upsieve-3.0.4 stack_data-0.6.3 terminado-0.18.1 tinycss2-1.5.1 tornado-6.6 traitlets-5.16.1 typing_extensions-4.16.0 tzdata-2026.5 uri-template-1.3.0 urllib
3-2.8.0 wcwidth-0.9.4 webcolors-25.10.0 webencodings-0.6.1 websocket-client-1.9.2
(.venv) PS D:\DTCC\GUVI\DevOps\python-da> python -c "import numpy, pandas; print('Numpy:', numpy.__version__); print('Pandas:', pandas.__version__)"
Numpy: 2.5.4
Pandas: 3.1.0
(.venv) PS D:\DTCC\GUVI\DevOps\python-da>
```

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Windows PowerShell
Downloading pure_eval-0.2.4-py3-none-any.whl (11 kB)
Downloading pycparser-3.12-py3-none-any.whl (62 kB)
Downloading rfc3339_validator-0.1.4-py2.py3-none-any.whl (3.5 kB)
Downloading tzdata-2026.5-py2.py3-none-any.whl (347 kB)
Downloading uri_template-1.3.0-py3-none-any.whl (11 kB)
Installing collected packages: pure-eval, websocket-client, webencodings, webcolors, wcwidth, urllib3, uri-template, tzdata, typing_extensions, traitlets, t
ornado, soupsieve, six, send2trash, rpds-py, rfc3986-validator, pyzmq, pyyaml, pywinpty, python-json-logger, pygments, pycparser, psutil, prometheus-client,
 platformdirs, parso, pandocfilters, packaging, numpy, nest-asyncio2, mistune, MarkupSafe, lark, jupyterlab-pygments, jsonpointer, json5, idna, h11, fqdn, f
astjsonschema, executing, defusedxml, debugpy, comm, colorama, charset_normalizer, certifi, babel, attrs, async-lru, asttokens, tinycss2, terminado, stack_d
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nts-lexers, httpcore, cffi, bleach, beautifulsoup4, anyio, pandas, jupyter-server-terminals, jupyter-client, jupyter-builder, jsonschema-specifications, ipy
thon, httpx, arrow, argon2-cffi-bindings, jsonschema, isoduration, ipykernel, argon2-cffi, nbformat, nbclient, jupyter-events, nbconvert, jupyter-server, no
tebook-shim, jupyterlab-server, jupyter-lsp, jupyterlab
Successfully installed MarkupSafe-3.0.4 anyio-4.15.1 argon2-cffi-25.1.0 argon2-cffi-bindings-26.1.0 arrow-1.4.0 asttokens-3.0.2 async-lru-2.4.0 attrs-26.1.0
 babel-2.18.0 beautifulsoup4-4.15.0 bleach-6.4.0 certifi-2026.10.23 cffi-2.1.1 charset_normalizer-3.5.2 colorama-0.4.6 comm-0.2.3 debugpy-1.8.22 defusedxml-
0.7.1 executing-2.3.0 fastjsonschema-2.22.2 fqdn-1.6.0 h11-0.16.0 httpcore-1.0.9 httpx-0.28.1 idna-3.20 ipykernel-7.4.0 ipython-9.17.1 ipython-pygments-lexe
rs-1.1.1 isoduration-20.11.0 jedi-0.20.1 jinja2-3.1.6 json5-0.17.3 jsonpointer-3.5 jsonschema-4.26.0 jsonschema-specifications-2025.9.1 jupyter-builder-1.2.
3 jupyter-client-8.10.0 jupyter-core-5.9.1 jupyter-events-0.12.1 jupyter-lsp-2.3.1 jupyter-server-2.21.1 jupyter-server-terminals-0.5.4 jupyterlab-4.6.4 jup
yterlab-pygments-0.3.0 jupyterlab-server-2.28.1 lark-1.3.1 matplotlib-inline-0.2.2 mistune-3.4.0 nbclient-0.11.0 nbconvert-7.17.2 nbformat-5.11.1 nest-async
io2-1.7.4 notebook-shim-0.2.4 numpy-2.5.4 packaging-26.3 pandas-3.1.0 pandocfilters-1.5.1 parso-0.8.7 platformdirs-4.13.5 prometheus-client-0.26.0 prompt_to
olkit-3.0.53 psutil-7.2.2 pure-eval-0.2.4 pycparser-3.12 pygments-2.21.0 python-dateutil-3.0.0 python-json-logger-4.2.0 pywinpty-3.1.0 pyyaml-6.0.3 pyzmq-27
.2.0 referencing-0.37.0 requests-2.34.2 rfc3339-validator-0.1.4 rfc3986-validator-0.1.1 rfc3987-syntax-1.1.0 rpds-py-2026.9.1 send2trash-2.1.0 six-1.17.0 so
upsieve-3.0.4 stack_data-0.6.3 terminado-0.18.1 tinycss2-1.5.1 tornado-6.6 traitlets-5.16.1 typing_extensions-4.16.0 tzdata-2026.5 uri-template-1.3.0 urllib
3-2.8.0 wcwidth-0.9.4 webcolors-25.10.0 webencodings-0.6.1 websocket-client-1.9.2
(.venv) PS D:\DTCC\GUVI\DevOps\python-da> python -c "import numpy, pandas; print('Numpy:', numpy.__version__); print('Pandas:', pandas.__version__)"
Numpy: 2.5.4
Pandas: 3.1.0
(.venv) PS D:\DTCC\GUVI\DevOps\python-da> Get-Content .\food_delivery.csv
Date,Product,Orders,Revenue,Delivery_Time,Rating,Status
2026-10-01,Burger,45,9000,25,4.5,Delivered
2026-10-01,Pizza,30,12000,35,4.7,Delivered
2026-10-01,Biryani,50,15000,40,4.2,Delivered
2026-10-02,Burger,50,10000,28,4.4,Delivered
2026-10-02,Pizza,35,14000,38,4.8,Delivered
2026-10-02,Biryani,55,16500,45,4.1,Late
2026-10-03,Sandwich,28,5600,21,4.1,Delivered
2026-10-03,Fried Rice,38,7600,31,4.4,Delivered
2026-10-04,Pizza,40,16000,42,4.6,Late
2026-10-04,Biryani,65,19500,48,4.3,Late
2026-10-05,Sandwich,35,7000,24,4.3,Delivered
2026-10-05,Fried Rice,42,8400,33,4.4,Delivered
(.venv) PS D:\DTCC\GUVI\DevOps\python-da>
```

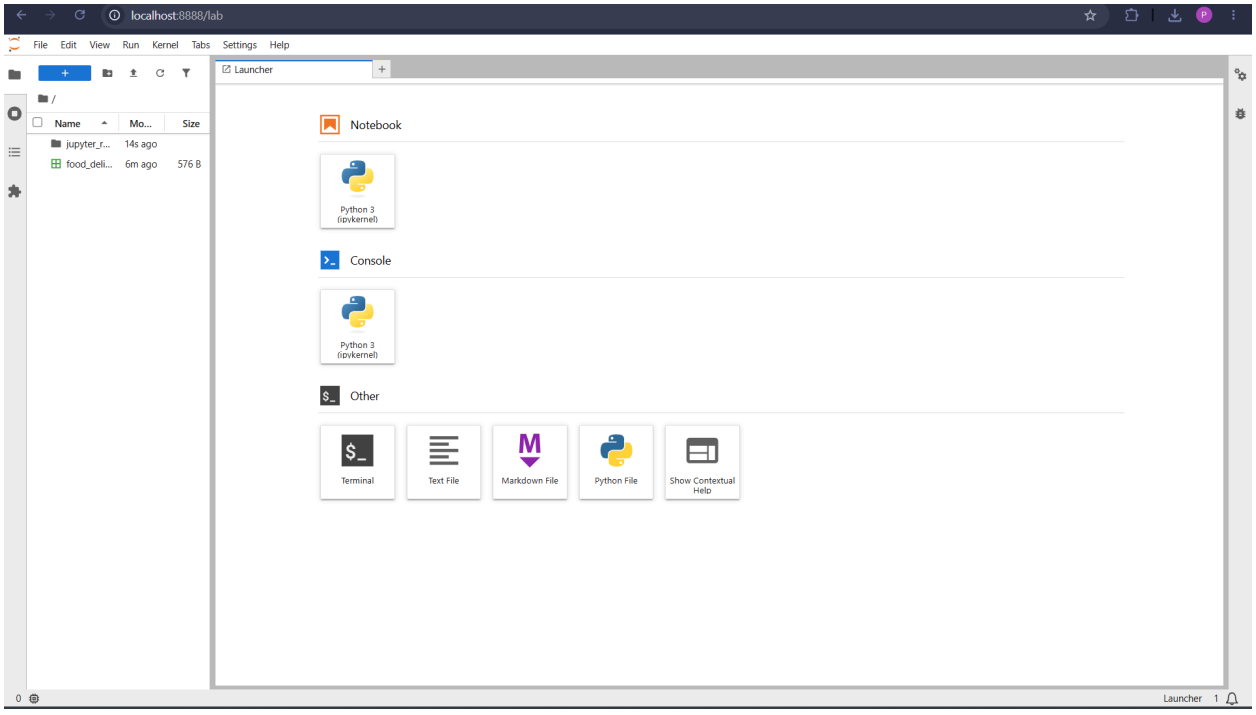
```
npm prefix
  File "D:\DTCC\GUVI\DevOps\python-da\.venv\Lib\site-packages\jupyter_server\serverapp.py", line 3039, in write_browser_open_file
    with open(self.browser_open_file, "w", encoding="utf-8") as f:
         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
PermissionError: [Errno 13] Permission denied: 'C:\\Users\\DELL\\AppData\\Roaming\\jupyter\\runtime\\jpserver-28164-open.html'
(.venv) PS D:\DTCC\GUVI\DevOps\python-da> mkdir D:\DTCC\GUVI\DevOps\python-da\jupyter_runtime -Force

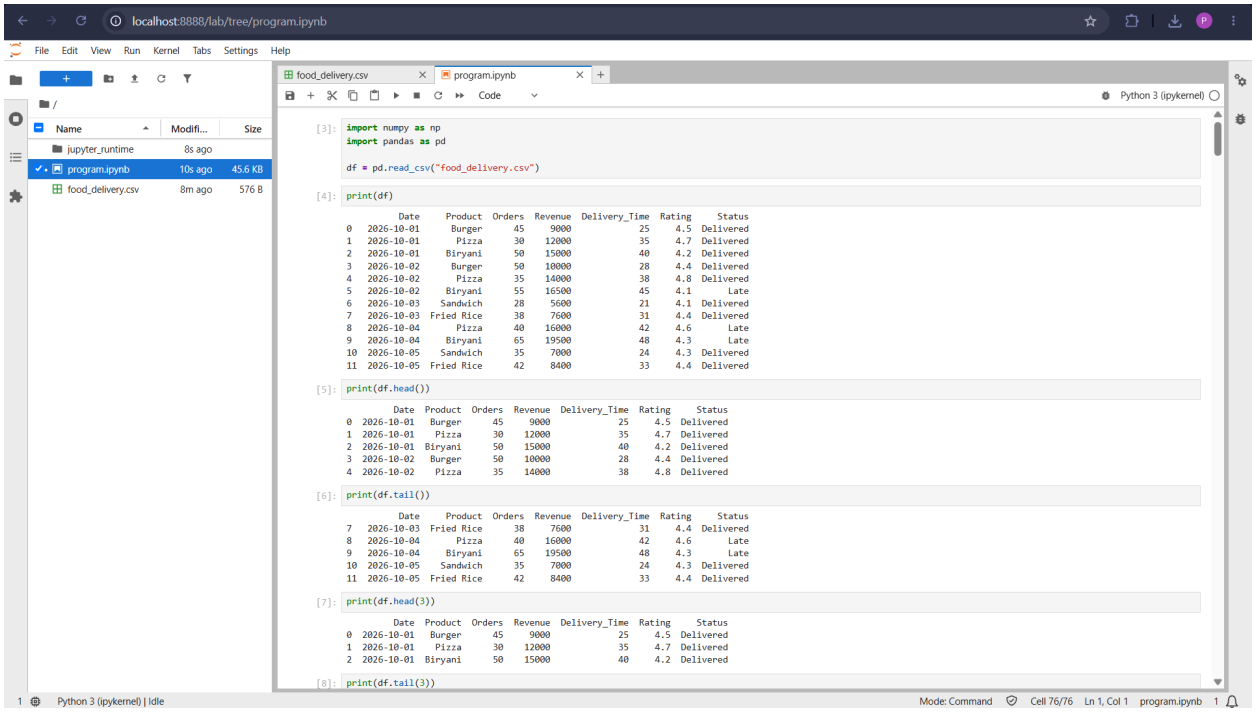

    Directory: D:\DTCC\GUVI\DevOps\python-da

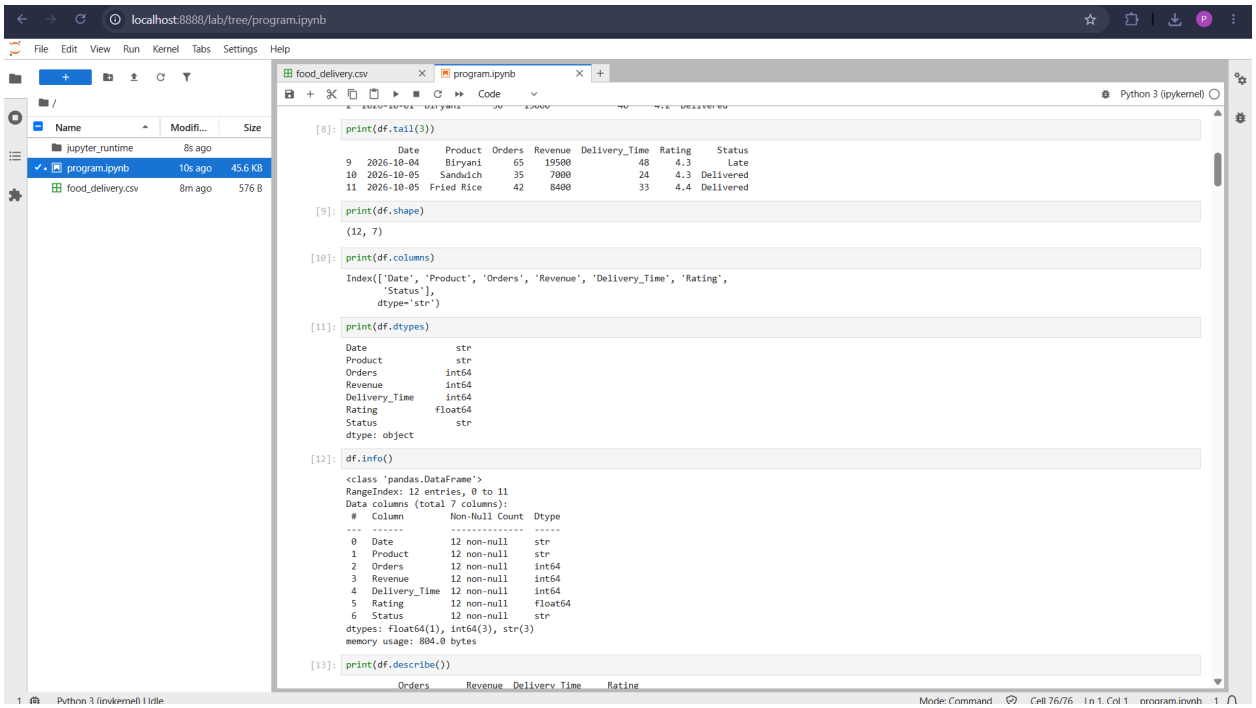

Mode                 LastWriteTime         Length Name
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d-----        26-10-2026     16:18                jupyter_runtime

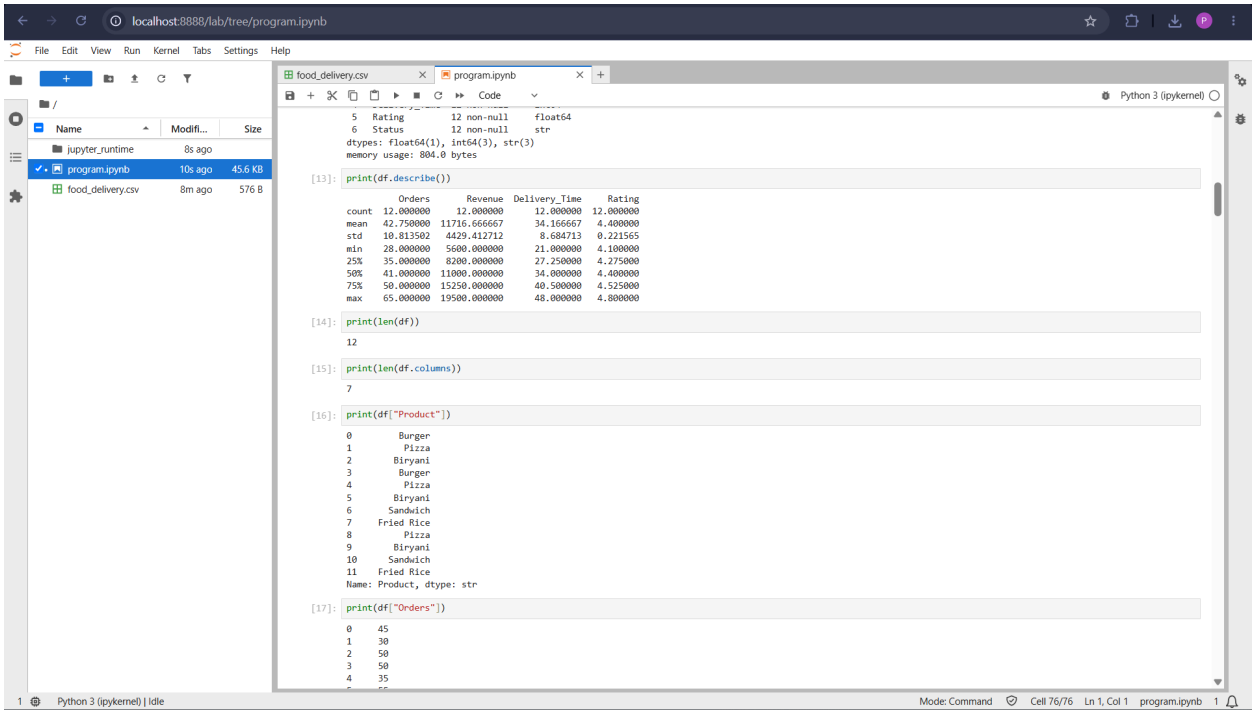

(.venv) PS D:\DTCC\GUVI\DevOps\python-da> $env:JUPYTER_RUNTIME_DIR = "D:\DTCC\GUVI\DevOps\python-da\jupyter_runtime"
(.venv) PS D:\DTCC\GUVI\DevOps\python-da> jupyter lab
[I 2026-10-26 16:19:06.628 ServerApp] jupyter_lsp | extension was successfully linked.
[I 2026-10-26 16:19:06.634 ServerApp] jupyter_server_terminals | extension was successfully linked.
[I 2026-10-26 16:19:06.641 ServerApp] jupyterlab | extension was successfully linked.
[I 2026-10-26 16:19:06.646 ServerApp] Writing Jupyter server cookie secret to D:\DTCC\GUVI\DevOps\python-da\jupyter_runtime\jupyter_cookie_secret
[I 2026-10-26 16:19:06.738 ServerApp] notebook_shim | extension was successfully linked.
[I 2026-10-26 16:19:06.830 ServerApp] notebook_shim | extension was successfully loaded.
[I 2026-10-26 16:19:06.833 ServerApp] jupyter_lsp | extension was successfully loaded.
[I 2026-10-26 16:19:06.833 ServerApp] jupyter_server_terminals | extension was successfully loaded.
[I 2026-10-26 16:19:06.837 LabApp] JupyterLab extension loaded from D:\DTCC\GUVI\DevOps\python-da\.venv\Lib\site-packages\jupyterlab
[I 2026-10-26 16:19:06.837 LabApp] JupyterLab application directory is D:\DTCC\GUVI\DevOps\python-da\.venv\share\jupyter\lab
[I 2026-10-26 16:19:06.838 LabApp] Extension Manager is 'pypi'.
[I 2026-10-26 16:19:07.084 ServerApp] jupyterlab | extension was successfully loaded.
[I 2026-10-26 16:19:07.086 ServerApp] Serving notebooks from local directory: D:\DTCC\GUVI\DevOps\python-da
[I 2026-10-26 16:19:07.086 ServerApp] Jupyter Server 2.21.1 is running at:
[I 2026-10-26 16:19:07.087 ServerApp] http://localhost:8888/lab?token=c85952a64b4748c42d5a6612f8d97fd90a3ac0dc47bbc77e
[I 2026-10-26 16:19:07.087 ServerApp]     http://127.0.0.1:8888/lab?token=c85952a64b4748c42d5a6612f8d97fd90a3ac0dc47bbc77e
[I 2026-10-26 16:19:07.087 ServerApp] Use Control-C to stop this server and shut down all kernels (twice to skip confirmation).
[C 2026-10-26 16:19:07.193 ServerApp]

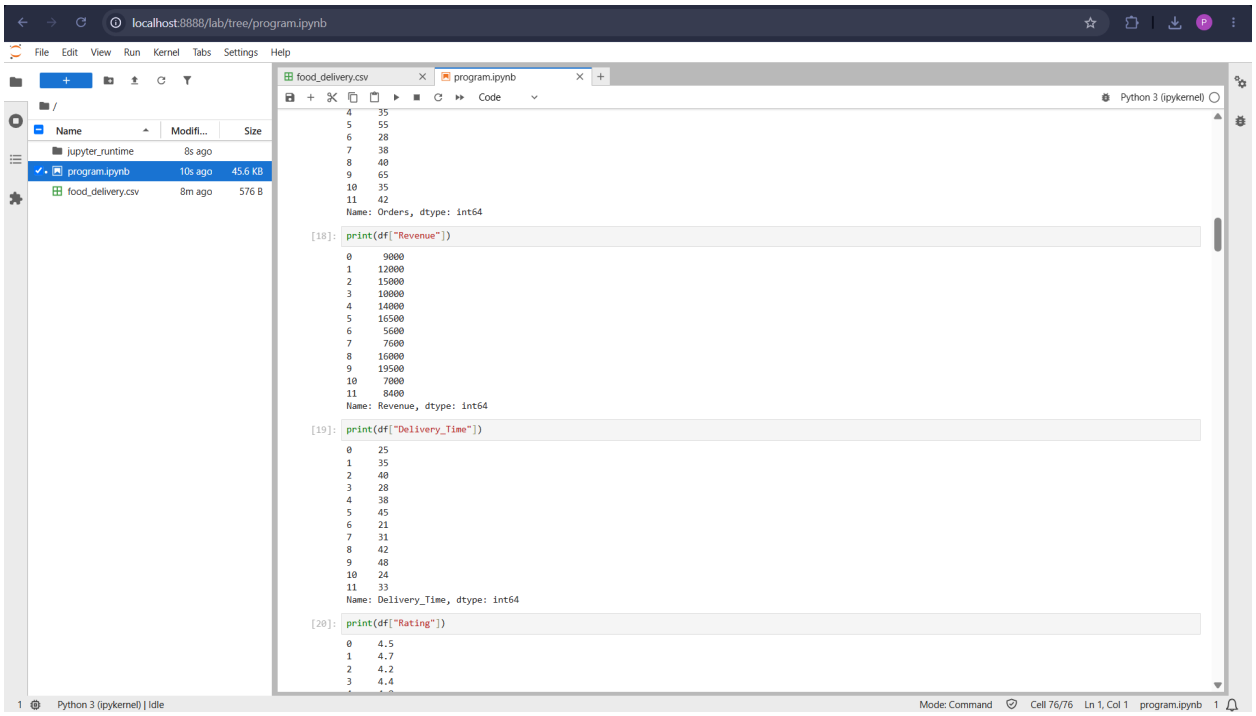
    To access the server, open this file in a browser:
        file:///D:/DTCC/GUVI/DevOps/python-da/jupyter_runtime/jpserver-9188-open.html
    Or copy and paste one of these URLs:
        http://localhost:8888/lab?token=c85952a64b4748c42d5a6612f8d97fd90a3ac0dc47bbc77e
        http://127.0.0.1:8888/lab?token=c85952a64b4748c42d5a6612f8d97fd90a3ac0dc47bbc77e
[I 2026-10-26 16:19:12.381 ServerApp] Skipped non-installed server(s): basedpyright, bash-language-server, dockerfile-language-server-nodejs, javascript-typescript-langserver, jedi-language-server, julia-language-server, pyrefly, pyright, python-language-server, python-lsp-server, r-languageserver, sql-language-server, texlab, typescript-language-server, unified-language-server, vscode-css-languageserver-bin, vscode-html-languageserver-bin, vscode-json-languageserver-bin, yaml-language-server
[I 2026-10-26 16:19:21.287 LabApp] Build is up to date
```

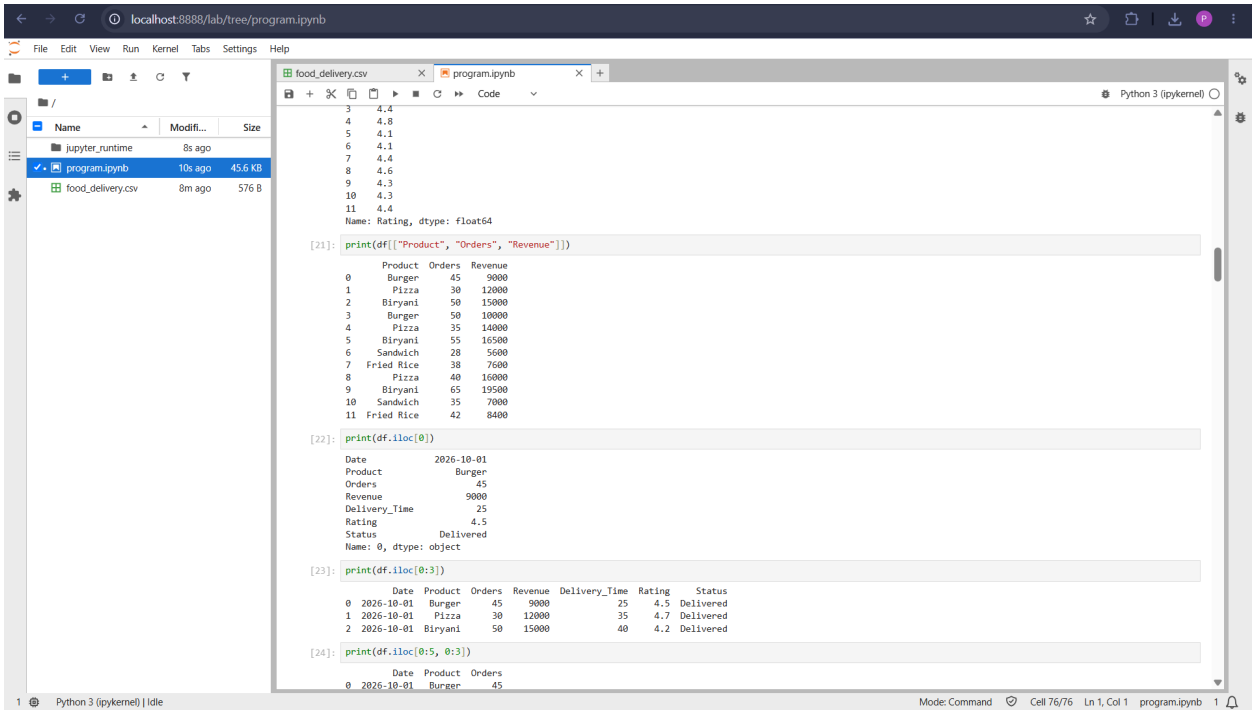


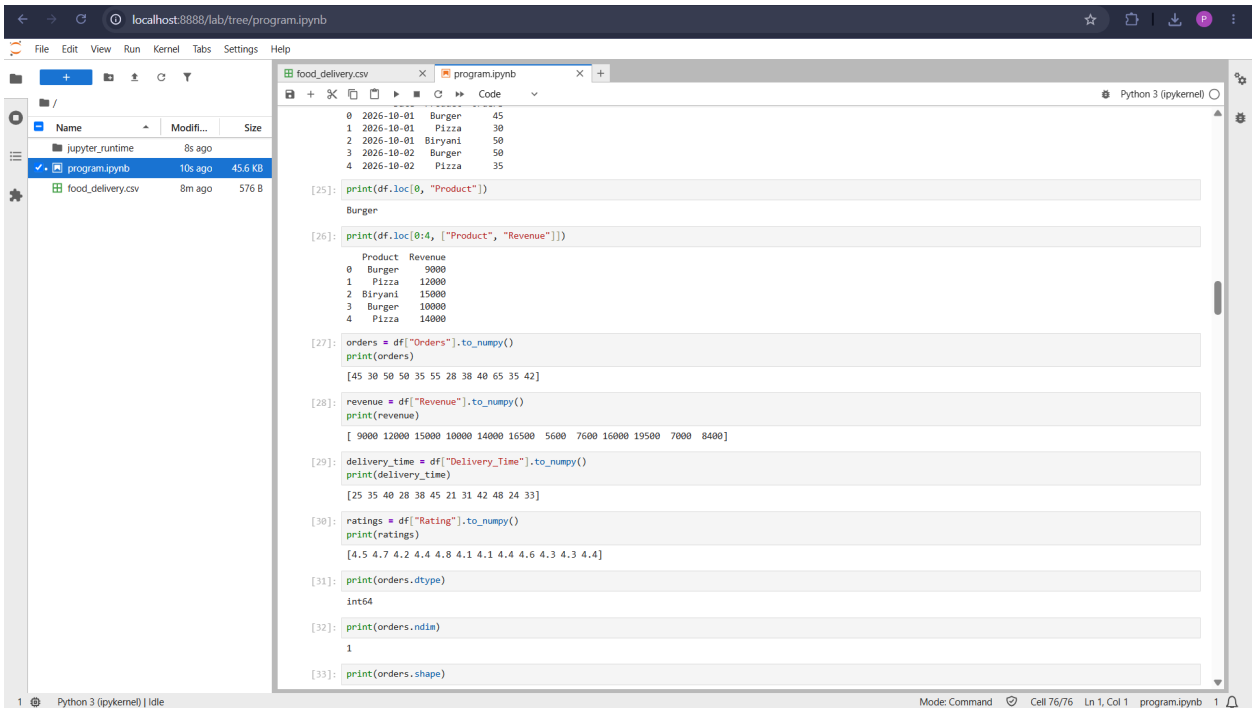

localhost:8888/lab/tree/program.ipynb
File Edit View Run Kernel Tabs Settings Help
food_delivery.csv
program.ipynb
Code
Python 3 (ipykernel)
Name
Modifi...
Size
jupyter_runtime 8s ago
program.ipynb 10s ago 45.6 KB
food_delivery.csv 8m ago 576 B
[3]: import numpy as np
import pandas as pd
df = pd.read_csv("food_delivery.csv")
[4]: print(df)
Date Product Orders Revenue Delivery_Time Rating Status
0 2026-10-01 Burger 45 9000 25 4.5 Delivered
1 2026-10-01 Pizza 30 12000 35 4.7 Delivered
2 2026-10-01 Biryani 50 15000 40 4.2 Delivered
3 2026-10-02 Burger 50 10000 28 4.4 Delivered
4 2026-10-02 Pizza 35 14000 38 4.8 Delivered
5 2026-10-02 Biryani 55 16500 45 4.1 Late
6 2026-10-03 Sandwich 28 5600 21 4.1 Delivered
7 2026-10-03 Fried Rice 38 7600 31 4.4 Delivered
8 2026-10-04 Pizza 40 16000 42 4.6 Late
9 2026-10-04 Biryani 65 19500 48 4.3 Late
10 2026-10-05 Sandwich 35 7000 24 4.3 Delivered
11 2026-10-05 Fried Rice 42 8400 33 4.4 Delivered
[5]: print(df.head())
Date Product Orders Revenue Delivery_Time Rating Status
0 2026-10-01 Burger 45 9000 25 4.5 Delivered
1 2026-10-01 Pizza 30 12000 35 4.7 Delivered
2 2026-10-01 Biryani 50 15000 40 4.2 Delivered
3 2026-10-02 Burger 50 10000 28 4.4 Delivered
4 2026-10-02 Pizza 35 14000 38 4.8 Delivered
[6]: print(df.tail())
Date Product Orders Revenue Delivery_Time Rating Status
7 2026-10-03 Fried Rice 38 7600 31 4.4 Delivered
8 2026-10-04 Pizza 40 16000 42 4.6 Late
9 2026-10-04 Biryani 65 19500 48 4.3 Late
10 2026-10-05 Sandwich 35 7000 24 4.3 Delivered
11 2026-10-05 Fried Rice 42 8400 33 4.4 Delivered
[7]: print(df.head(3))
Date Product Orders Revenue Delivery_Time Rating Status
0 2026-10-01 Burger 45 9000 25 4.5 Delivered
1 2026-10-01 Pizza 30 12000 35 4.7 Delivered
2 2026-10-01 Biryani 50 15000 40 4.2 Delivered
[8]: print(df.tail(3))
Python 3 (ipykernel) | Idle
Mode: Command Cell 76/76 Ln 1, Col 1 program.ipynb

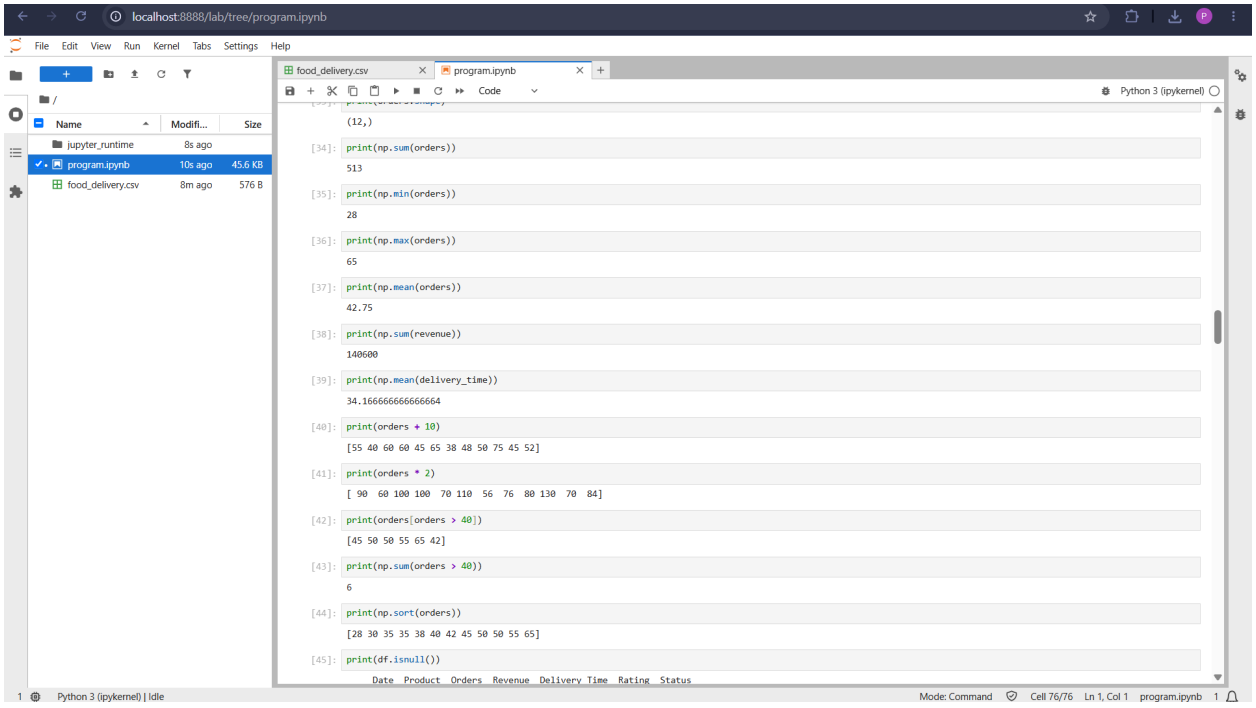

localhost:8888/lab/tree/program.ipynb
File Edit View Run Kernel Tabs Settings Help
food_delivery.csv
program.ipynb
Code
Python 3 (ipykernel)
Name
Modifi...
Size
jupyter_runtime 8s ago
program.ipynb 10s ago 45.6 KB
food_delivery.csv 8m ago 576 B
[8]: print(df.tail(3))
Date Product Orders Revenue Delivery_Time Rating Status
9 2026-10-04 Biryani 65 19500 48 4.3 Late
10 2026-10-05 Sandwich 35 7000 24 4.3 Delivered
11 2026-10-05 Fried Rice 42 8400 33 4.4 Delivered
[9]: print(df.shape)
(12, 7)
[10]: print(df.columns)
Index(['Date', 'Product', 'Orders', 'Revenue', 'Delivery_Time', 'Rating',
'Status'],
dtype='str')
[11]: print(df.dtypes)
Date str
Product str
Orders int64
Revenue int64
Delivery_Time int64
Rating float64
Status str
dtype: object
[12]: df.info()
<class 'pandas.DataFrame'>
RangeIndex: 12 entries, 0 to 11
Data columns (total 7 columns):
# Column Non-Null Count Dtype
0 Date 12 non-null str
1 Product 12 non-null str
2 Orders 12 non-null int64
3 Revenue 12 non-null int64
4 Delivery_Time 12 non-null int64
5 Rating 12 non-null float64
6 Status 12 non-null str
dtypes: float64(1), int64(3), str(3)
memory usage: 804.0 bytes
[13]: print(df.describe())
Orders Revenue Delivery_Time Rating
Mode: Command Cell 76/76 Ln 1, Col 1 program.ipynb

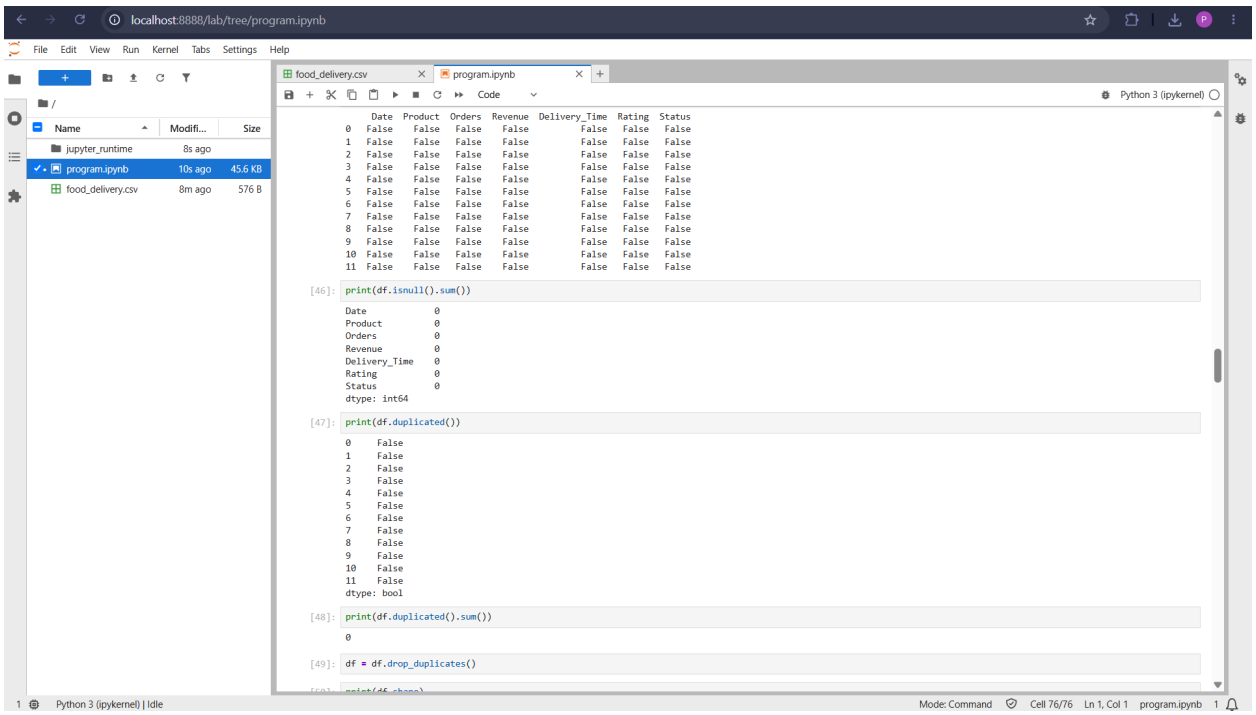

localhost:8888/lab/tree/program.ipynb
File Edit View Run Kernel Tabs Settings Help
food_delivery.csv
program.ipynb
Code
Python 3 (ipykernel)
Name
Modifi...
Size
jupyter_runtime 8s ago
program.ipynb 10s ago 45.6 KB
food_delivery.csv 8m ago 576 B
5 Rating 12 non-null float64
6 Status 12 non-null str
dtypes: float64(1), int64(3), str(3)
memory usage: 804.0 bytes
[13]: print(df.describe())
Orders Revenue Delivery_Time Rating
count 12.000000 12.000000 12.000000 12.000000
mean 42.750000 11716.666667 34.166667 4.400000
std 10.813502 4429.412712 8.684713 0.221565
min 28.000000 5600.000000 21.000000 4.100000
25% 35.000000 8200.000000 27.250000 4.275000
50% 41.000000 11000.000000 34.000000 4.400000
75% 50.000000 15250.000000 40.500000 4.525000
max 65.000000 19500.000000 48.000000 4.800000
[14]: print(len(df))
12
[15]: print(len(df.columns))
7
[16]: print(df["Product"])
0 Burger
1 Pizza
2 Biryani
3 Burger
4 Pizza
5 Biryani
6 Sandwich
7 Fried Rice
8 Pizza
9 Biryani
10 Sandwich
11 Fried Rice
Name: Product, dtype: str
[17]: print(df["Orders"])
0 45
1 30
2 50
3 50
4 35
Python 3 (ipykernel) | Idle
Mode: Command
Cell 76/76 Ln 1, Col 1 program.ipynb

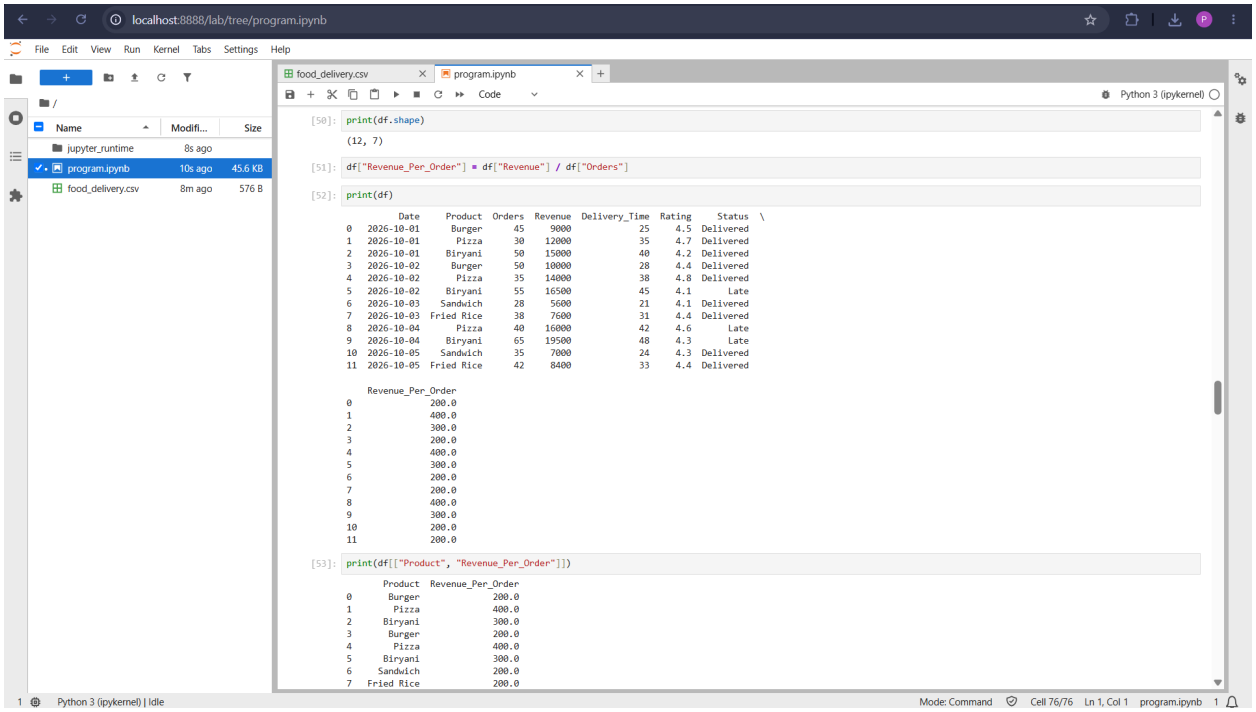

localhost:8888/lab/tree/program.ipynb
File Edit View Run Kernel Tabs Settings Help
food_delivery.csv
program.ipynb
Code
Python 3 (ipykernel)
Name
Modifi...
Size
jupyter_runtime 8s ago
program.ipynb 10s ago 45.6 KB
food_delivery.csv 8m ago 576 B
4 35
5 55
6 28
7 38
8 40
9 65
10 35
11 42
Name: Orders, dtype: int64
[18]: print(df["Revenue"])
0 9000
1 12000
2 15000
3 10000
4 14000
5 16500
6 5600
7 7600
8 16000
9 19500
10 7000
11 8400
Name: Revenue, dtype: int64
[19]: print(df["Delivery_Time"])
0 25
1 35
2 40
3 28
4 38
5 45
6 21
7 31
8 42
9 48
10 24
11 33
Name: Delivery_Time, dtype: int64
[20]: print(df["Rating"])
0 4.5
1 4.7
2 4.2
3 4.4
Python 3 (ipykernel) | Idle
Mode: Command
Cell 76/76 Ln 1, Col 1 program.ipynb

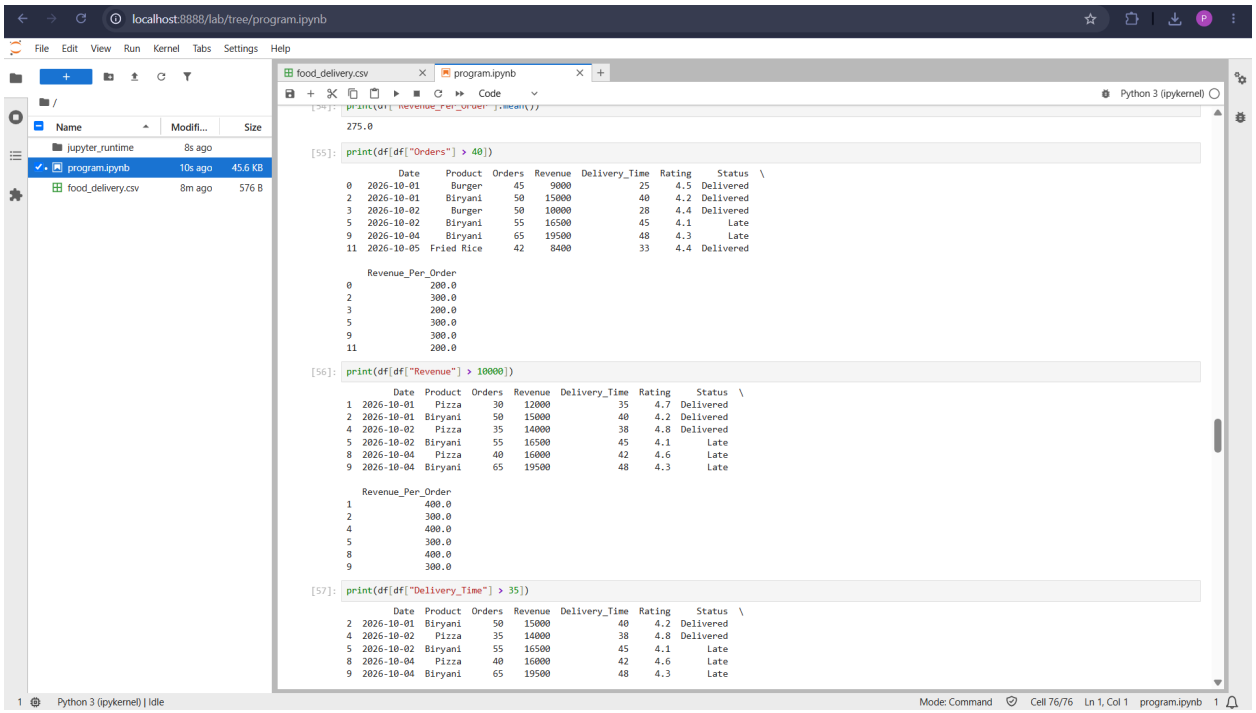

localhost:8888/lab/tree/program.ipynb
File Edit View Run Kernel Tabs Settings Help
food_delivery.csv
program.ipynb
Code
Python 3 (ipykernel)
Name
Modifi...
Size
jupyter_runtime 8s ago
program.ipynb 10s ago 45.6 KB
food_delivery.csv 8m ago 576 B
3 4.4
4 4.8
5 4.1
6 4.1
7 4.4
8 4.6
9 4.3
10 4.3
11 4.4
Name: Rating, dtype: float64
[21]: print(df[["Product", "Orders", "Revenue"]])
Product Orders Revenue
0 Burger 45 9000
1 Pizza 30 12000
2 Biryani 50 15000
3 Burger 50 10000
4 Pizza 35 14000
5 Biryani 55 16500
6 Sandwich 28 5600
7 Fried Rice 38 7600
8 Pizza 40 16000
9 Biryani 65 19500
10 Sandwich 35 7000
11 Fried Rice 42 8400
[22]: print(df.iloc[0])
Date 2026-10-01
Product Burger
Orders 45
Revenue 9000
Delivery_Time 25
Rating 4.5
Status Delivered
Name: 0, dtype: object
[23]: print(df.iloc[0:3])
Date Product Orders Revenue Delivery_Time Rating Status
0 2026-10-01 Burger 45 9000 25 4.5 Delivered
1 2026-10-01 Pizza 30 12000 35 4.7 Delivered
2 2026-10-01 Biryani 50 15000 40 4.2 Delivered
[24]: print(df.iloc[0:5, 0:3])
Date Product Orders
0 2026-10-01 Burger 45
Python 3 (ipykernel) | Idle
Mode: Command
Cell 76/76
Ln 1, Col 1
program.ipynb

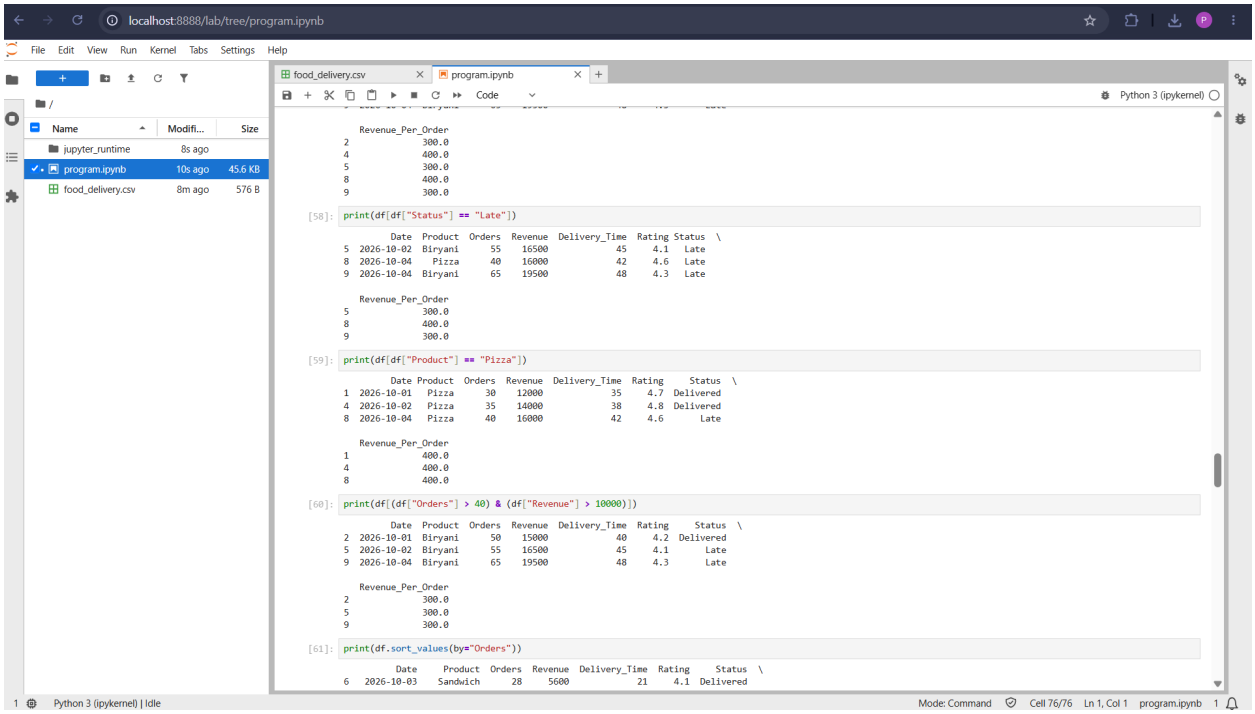

localhost:8888/lab/tree/program.ipynb
File Edit View Run Kernel Tabs Settings Help
food_delivery.csv
program.ipynb
Code
Python 3 (ipykernel)
Name
Modifi...
Size
jupyter_runtime 8s ago
program.ipynb 10s ago 45.6 KB
food_delivery.csv 8m ago 576 B
0 2026-10-01 Burger 45
1 2026-10-01 Pizza 30
2 2026-10-01 Biryani 50
3 2026-10-02 Burger 50
4 2026-10-02 Pizza 35
[25]: print(df.loc[0, "Product"])
Burger
[26]: print(df.loc[0:4, ["Product", "Revenue"]])
Product Revenue
0 Burger 9000
1 Pizza 12000
2 Biryani 15000
3 Burger 10000
4 Pizza 14000
[27]: orders = df["Orders"].to_numpy()
print(orders)
[45 30 50 50 35 55 28 38 40 65 35 42]
[28]: revenue = df["Revenue"].to_numpy()
print(revenue)
[ 9000 12000 15000 10000 14000 16500 5600 7600 16000 19500 7000 8400]
[29]: delivery_time = df["Delivery_Time"].to_numpy()
print(delivery_time)
[25 35 40 28 38 45 21 31 42 48 24 33]
[30]: ratings = df["Rating"].to_numpy()
print(ratings)
[4.5 4.7 4.2 4.4 4.8 4.1 4.1 4.4 4.6 4.3 4.3 4.4]
[31]: print(orders.dtype)
int64
[32]: print(orders.ndim)
1
[33]: print(orders.shape)
Python 3 (ipykernel) | Idle
Mode: Command
Cell 76/76
Ln 1, Col 1
program.ipynb

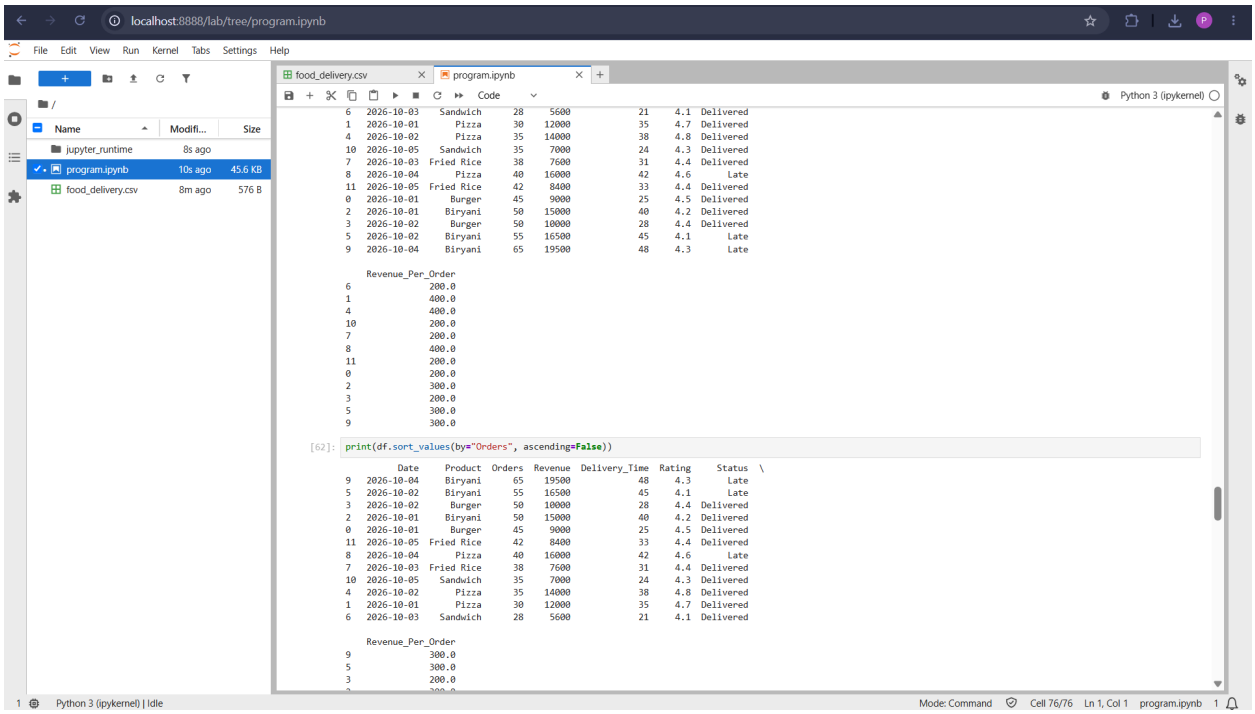

localhost:8888/lab/tree/program.ipynb
File Edit View Run Kernel Tabs Settings Help
food_delivery.csv
program.ipynb
Code
Python 3 (ipykernel)
Name
Modifi...
Size
jupyter_runtime 8s ago
program.ipynb 10s ago 45.6 KB
food_delivery.csv 8m ago 576 B
(12,)
[34]: print(np.sum(orders))
513
[35]: print(np.min(orders))
28
[36]: print(np.max(orders))
65
[37]: print(np.mean(orders))
42.75
[38]: print(np.sum(revenue))
140600
[39]: print(np.mean(delivery_time))
34.166666666666664
[40]: print(orders + 10)
[55 40 60 60 45 65 38 48 50 75 45 52]
[41]: print(orders * 2)
[ 90 60 100 100 70 110 56 76 80 130 70 84]
[42]: print(orders[orders > 40])
[45 50 50 55 65 42]
[43]: print(np.sum(orders > 40))
6
[44]: print(np.sort(orders))
[28 30 35 35 38 40 42 45 50 50 55 65]
[45]: print(df.isnull())
Date Product Orders Revenue Delivery_Time Rating Status
Python 3 (ipykernel) | Idle
Mode: Command
Cell 76/76
Ln 1, Col 1
program.ipynb

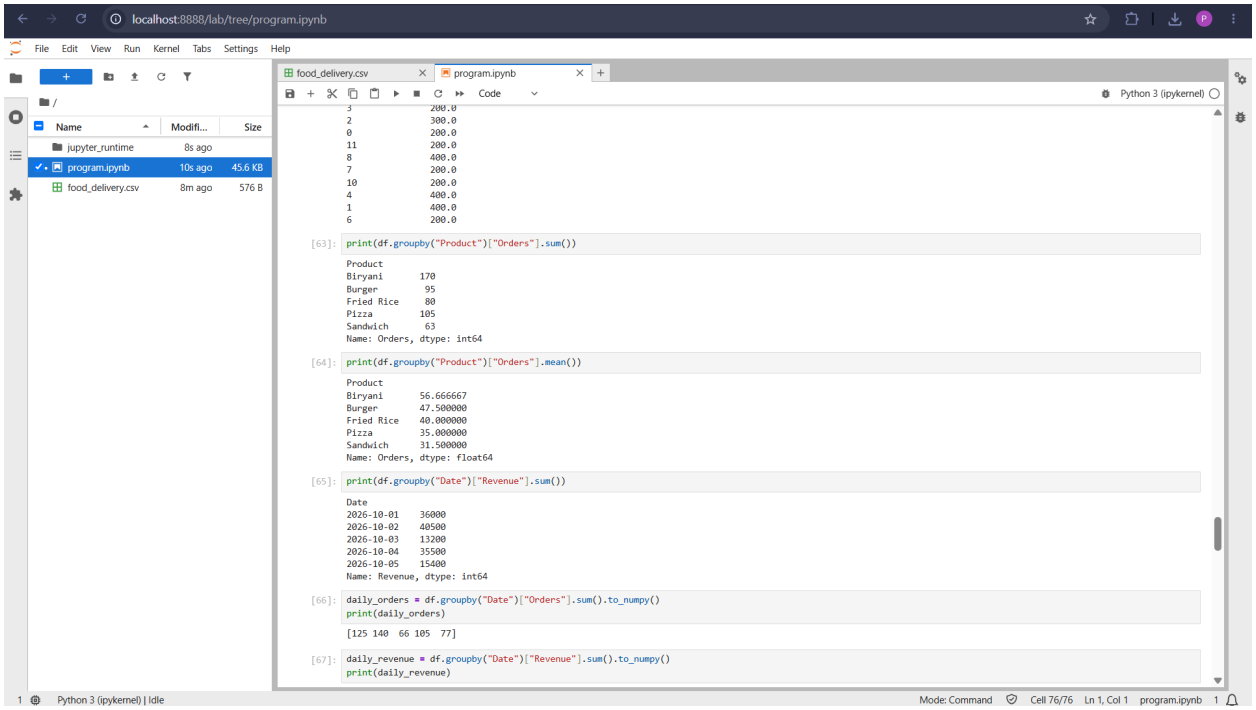

localhost:8888/lab/tree/program.ipynb
File Edit View Run Kernel Tabs Settings Help
food_delivery.csv
program.ipynb
Code
Python 3 (ipykernel)
Name
Modifi...
Size
jupyter_runtime 8s ago
program.ipynb 10s ago 45.6 KB
food_delivery.csv 8m ago 576 B
Date Product Orders Revenue Delivery_Time Rating Status
0 False False False False False False False
1 False False False False False False False
2 False False False False False False False
3 False False False False False False False
4 False False False False False False False
5 False False False False False False False
6 False False False False False False False
7 False False False False False False False
8 False False False False False False False
9 False False False False False False False
10 False False False False False False False
11 False False False False False False False
[46]: print(df.isnull().sum())
Date 0
Product 0
Orders 0
Revenue 0
Delivery_Time 0
Rating 0
Status 0
dtype: int64
[47]: print(df.duplicated())
0 False
1 False
2 False
3 False
4 False
5 False
6 False
7 False
8 False
9 False
10 False
11 False
dtype: bool
[48]: print(df.duplicated().sum())
0
[49]: df = df.drop_duplicates()
Python 3 (ipykernel) | Idle
Mode: Command
Cell 76/76
Ln 1, Col 1
program.ipynb

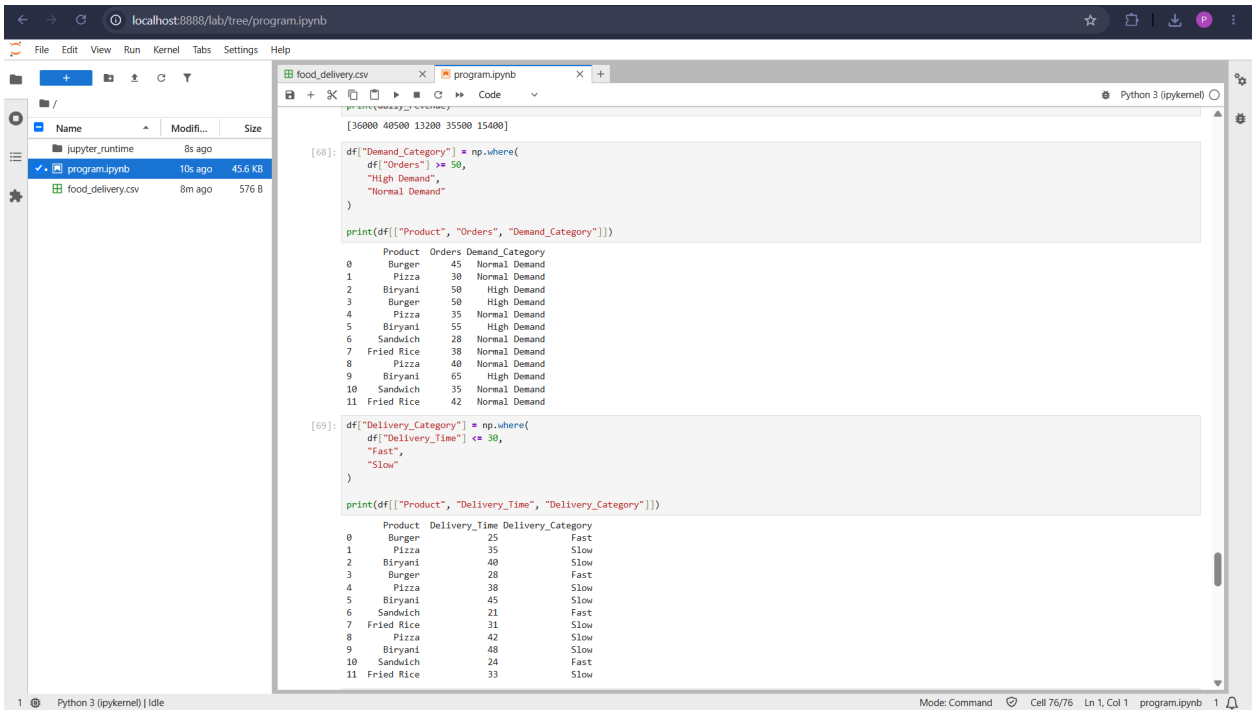

localhost:8888/lab/tree/program.ipynb
File Edit View Run Kernel Tabs Settings Help
food_delivery.csv
program.ipynb
Code
Python 3 (ipykernel)
Name
Modifi...
Size
jupyter_runtime 8s ago
program.ipynb 10s ago 45.6 KB
food_delivery.csv 8m ago 576 B
[50]: print(df.shape)
(12, 7)
[51]: df["Revenue_Per_Order"] = df["Revenue"] / df["Orders"]
[52]: print(df)
Date Product Orders Revenue Delivery_Time Rating Status \
0 2026-10-01 Burger 45 9000 25 4.5 Delivered
1 2026-10-01 Pizza 30 12000 35 4.7 Delivered
2 2026-10-01 Biryani 50 15000 40 4.2 Delivered
3 2026-10-02 Burger 50 10000 28 4.4 Delivered
4 2026-10-02 Pizza 35 14000 38 4.8 Delivered
5 2026-10-02 Biryani 55 16500 45 4.1 Late
6 2026-10-03 Sandwich 28 5600 21 4.1 Delivered
7 2026-10-03 Fried Rice 38 7600 31 4.4 Delivered
8 2026-10-04 Pizza 40 16000 42 4.6 Late
9 2026-10-04 Biryani 65 19500 48 4.3 Late
10 2026-10-05 Sandwich 35 7000 24 4.3 Delivered
11 2026-10-05 Fried Rice 42 8400 33 4.4 Delivered
Revenue_Per_Order
0 200.0
1 400.0
2 300.0
3 200.0
4 400.0
5 300.0
6 200.0
7 200.0
8 400.0
9 300.0
10 200.0
11 200.0
[53]: print(df[["Product", "Revenue_Per_Order"]])
Product Revenue_Per_Order
0 Burger 200.0
1 Pizza 400.0
2 Biryani 300.0
3 Burger 200.0
4 Pizza 400.0
5 Biryani 300.0
6 Sandwich 200.0
7 Fried Rice 200.0
Python 3 (ipykernel) | Idle
Mode: Command Cell 76/76 Ln 1, Col 1 program.ipynb

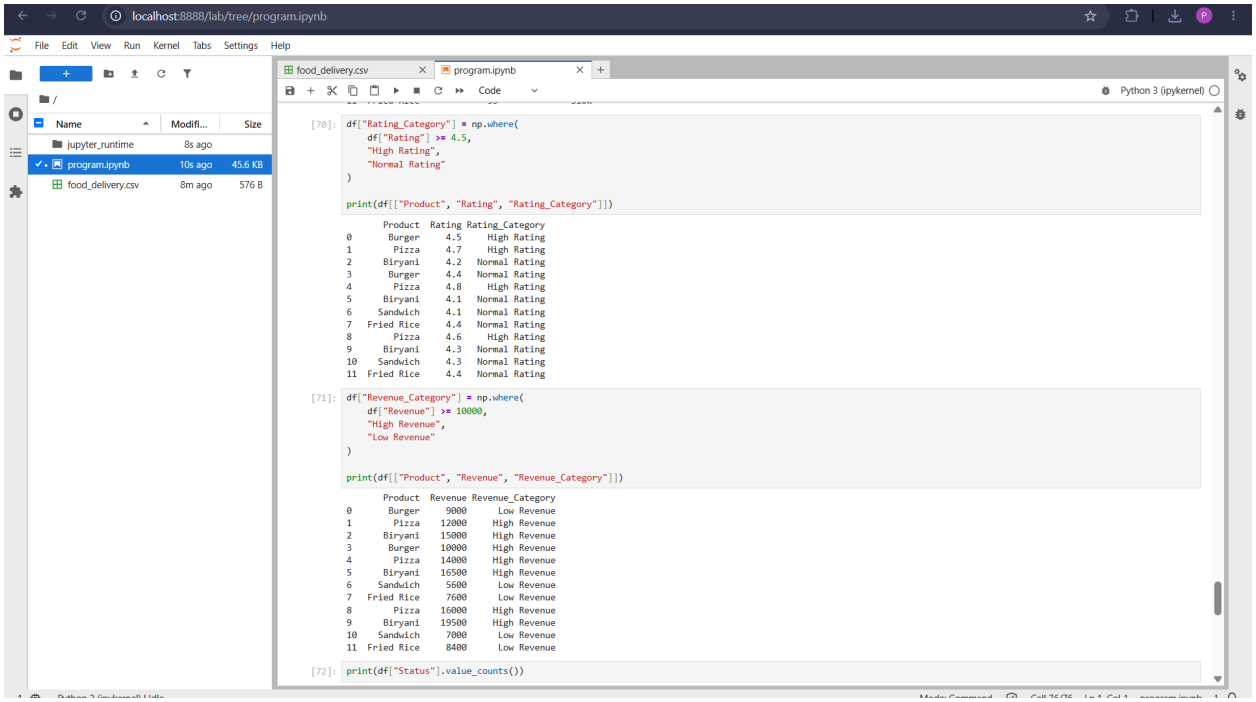

localhost:8888/lab/tree/program.ipynb
File Edit View Run Kernel Tabs Settings Help
food_delivery.csv
program.ipynb
Code
Python 3 (ipykernel)
Name
Modifi...
Size
jupyter_runtime 8s ago
program.ipynb 10s ago 45.6 KB
food_delivery.csv 8m ago 576 B
275.0
[55]: print(df[df["Orders"] > 40])
Date Product Orders Revenue Delivery_Time Rating Status \
0 2026-10-01 Burger 45 9000 25 4.5 Delivered
2 2026-10-01 Biryani 50 15000 40 4.2 Delivered
3 2026-10-02 Burger 50 10000 28 4.4 Delivered
5 2026-10-02 Biryani 55 16500 45 4.1 Late
9 2026-10-04 Biryani 65 19500 48 4.3 Late
11 2026-10-05 Fried Rice 42 8400 33 4.4 Delivered
Revenue_Per_Order
0 200.0
2 300.0
3 200.0
5 300.0
9 300.0
11 200.0
[56]: print(df[df["Revenue"] > 10000])
Date Product Orders Revenue Delivery_Time Rating Status \
1 2026-10-01 Pizza 30 12000 35 4.7 Delivered
2 2026-10-01 Biryani 50 15000 40 4.2 Delivered
4 2026-10-02 Pizza 35 14000 38 4.8 Delivered
5 2026-10-02 Biryani 55 16500 45 4.1 Late
8 2026-10-04 Pizza 40 16000 42 4.6 Late
9 2026-10-04 Biryani 65 19500 48 4.3 Late
Revenue_Per_Order
1 400.0
2 300.0
4 400.0
5 300.0
8 400.0
9 300.0
[57]: print(df[df["Delivery_Time"] > 35])
Date Product Orders Revenue Delivery_Time Rating Status \
2 2026-10-01 Biryani 50 15000 40 4.2 Delivered
4 2026-10-02 Pizza 35 14000 38 4.8 Delivered
5 2026-10-02 Biryani 55 16500 45 4.1 Late
8 2026-10-04 Pizza 40 16000 42 4.6 Late
9 2026-10-04 Biryani 65 19500 48 4.3 Late
Python 3 (ipykernel) | Idle
Mode: Command Cell 76/76 Ln 1, Col 1 program.ipynb

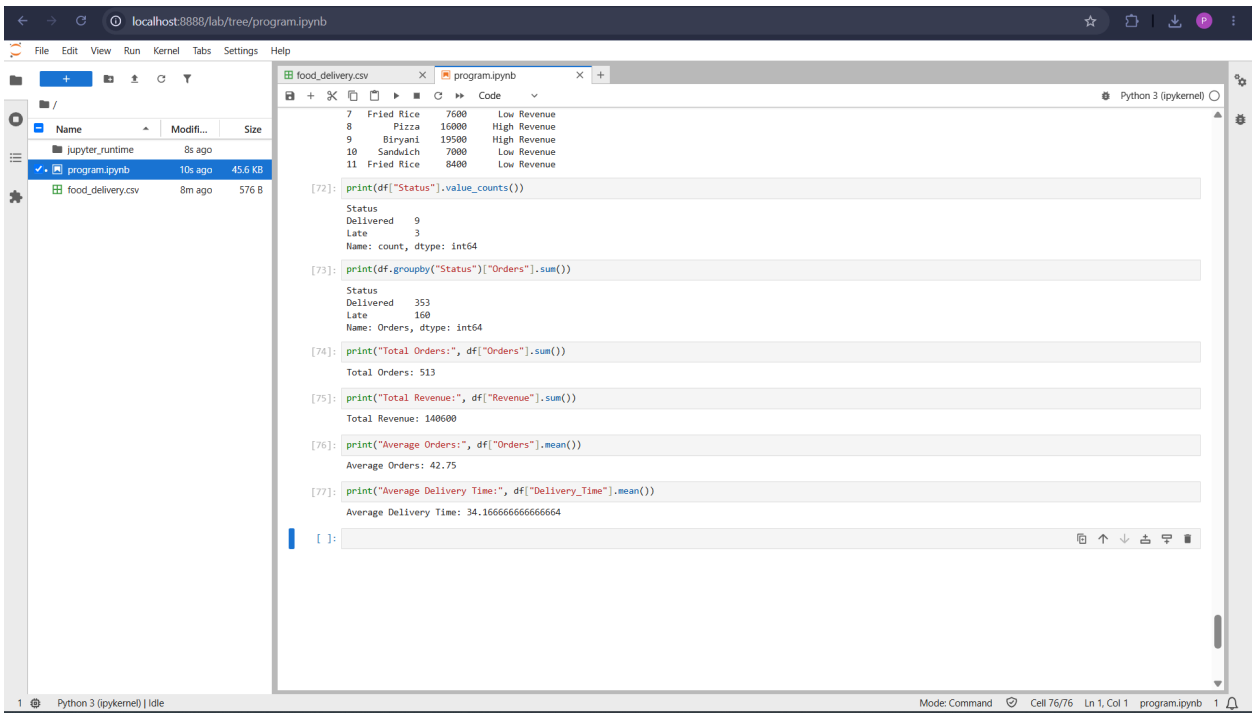

localhost:8888/lab/tree/program.ipynb
File Edit View Run Kernel Tabs Settings Help
food_delivery.csv
program.ipynb
Code
Python 3 (ipykernel)
Name
Modifi...
Size
jupyter_runtime 8s ago
program.ipynb 10s ago 45.6 KB
food_delivery.csv 8m ago 576 B
[58]: print(df[df["Status"] == "Late"])
[59]: print(df[df["Product"] == "Pizza"])
[60]: print(df[(df["Orders"] > 40) & (df["Revenue"] > 10000)])
[61]: print(df.sort_values(by="Orders"))
Python 3 (ipykernel) | Idle
Mode: Command
Cell 76/76
Ln 1, Col 1
program.ipynb


localhost:8888/lab/tree/program.ipynb
File Edit View Run Kernel Tabs Settings Help
food_delivery.csv
program.ipynb
Code
Python 3 (ipykernel)
Name
Modifi...
Size
jupyter_runtime 8s ago
program.ipynb 10s ago 45.6 KB
food_delivery.csv 8m ago 576 B
[62]: print(df.sort_values(by="Orders", ascending=False))
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3 200.0
2 300.0
0 200.0
11 200.0
8 400.0
7 200.0
10 200.0
4 400.0
1 400.0
6 200.0
[63]: print(df.groupby("Product")["Orders"].sum())
Product
Biryani 170
Burger 95
Fried Rice 80
Pizza 105
Sandwich 63
Name: Orders, dtype: int64
[64]: print(df.groupby("Product")["Orders"].mean())
Product
Biryani 56.666667
Burger 47.500000
Fried Rice 40.000000
Pizza 35.000000
Sandwich 31.500000
Name: Orders, dtype: float64
[65]: print(df.groupby("Date")["Revenue"].sum())
Date
2026-10-01 36000
2026-10-02 40500
2026-10-03 13200
2026-10-04 35500
2026-10-05 15400
Name: Revenue, dtype: int64
[66]: daily_orders = df.groupby("Date")["Orders"].sum().to_numpy()
print(daily_orders)
[125 140 66 105 77]
[67]: daily_revenue = df.groupby("Date")["Revenue"].sum().to_numpy()
print(daily_revenue)
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[36000 40500 13200 35500 15400]
[68]: df["Demand_Category"] = np.where(
df["Orders"] >= 50,
"High Demand",
"Normal Demand"
)
print(df[["Product", "Orders", "Demand_Category"]])
Product Orders Demand_Category
0 Burger 45 Normal Demand
1 Pizza 30 Normal Demand
2 Biryani 50 High Demand
3 Burger 50 High Demand
4 Pizza 35 Normal Demand
5 Biryani 55 High Demand
6 Sandwich 28 Normal Demand
7 Fried Rice 38 Normal Demand
8 Pizza 40 Normal Demand
9 Biryani 65 High Demand
10 Sandwich 35 Normal Demand
11 Fried Rice 42 Normal Demand
[69]: df["Delivery_Category"] = np.where(
df["Delivery_Time"] <= 30,
"Fast",
"Slow"
)
print(df[["Product", "Delivery_Time", "Delivery_Category"]])
Product Delivery_Time Delivery_Category
0 Burger 25 Fast
1 Pizza 35 Slow
2 Biryani 40 Slow
3 Burger 28 Fast
4 Pizza 38 Slow
5 Biryani 45 Slow
6 Sandwich 21 Fast
7 Fried Rice 31 Slow
8 Pizza 42 Slow
9 Biryani 48 Slow
10 Sandwich 24 Fast
11 Fried Rice 33 Slow
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[70]:
df["Rating_Category"] = np.where(
df["Rating"] >= 4.5,
"High Rating",
"Normal Rating"
)
print(df[["Product", "Rating", "Rating_Category"]])
Product Rating Rating_Category
0 Burger 4.5 High Rating
1 Pizza 4.7 High Rating
2 Biryani 4.2 Normal Rating
3 Burger 4.4 Normal Rating
4 Pizza 4.8 High Rating
5 Biryani 4.1 Normal Rating
6 Sandwich 4.1 Normal Rating
7 Fried Rice 4.4 Normal Rating
8 Pizza 4.6 High Rating
9 Biryani 4.3 Normal Rating
10 Sandwich 4.3 Normal Rating
11 Fried Rice 4.4 Normal Rating
[71]:
df["Revenue_Category"] = np.where(
df["Revenue"] >= 10000,
"High Revenue",
"Low Revenue"
)
print(df[["Product", "Revenue", "Revenue_Category"]])
Product Revenue Revenue_Category
0 Burger 9000 Low Revenue
1 Pizza 12000 High Revenue
2 Biryani 15000 High Revenue
3 Burger 10000 High Revenue
4 Pizza 14000 High Revenue
5 Biryani 16500 High Revenue
6 Sandwich 5600 Low Revenue
7 Fried Rice 7600 Low Revenue
8 Pizza 16000 High Revenue
9 Biryani 19500 High Revenue
10 Sandwich 7000 Low Revenue
11 Fried Rice 8400 Low Revenue
[72]:
print(df["Status"].value_counts())


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7 Fried Rice 7600 Low Revenue
8 Pizza 16000 High Revenue
9 Biryani 19500 High Revenue
10 Sandwich 7000 Low Revenue
11 Fried Rice 8400 Low Revenue
[72]:
print(df["Status"].value_counts())
Status
Delivered 9
Late 3
Name: count, dtype: int64
[73]:
print(df.groupby("Status")["Orders"].sum())
Status
Delivered 353
Late 160
Name: Orders, dtype: int64
[74]:
print("Total Orders:", df["Orders"].sum())
Total Orders: 513
[75]:
print("Total Revenue:", df["Revenue"].sum())
Total Revenue: 140600
[76]:
print("Average Orders:", df["Orders"].mean())
Average Orders: 42.75
[77]:
print("Average Delivery Time:", df["Delivery_Time"].mean())
Average Delivery Time: 34.166666666666664
[ ]:
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